Synapta: Prompt-Level Multi-Adapter LoRA Composition with Confidence-Gated Routing for On-Device LLM Inference

A Pre-Registered Empirical Study on Apple Silicon UMA

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All reported numbers derive from real model inference on Apple Silicon UMA. Total: 800+ inferences across Qwen2.5-1.5B-Instruct-4bit and Mistral-7B-Instruct-v0.3-4bit.

Abstract

I investigate whether multiple domain-specific LoRA adapters can be composed at inference time to handle genuinely multi-domain queries on edge hardware — specifically Apple Silicon Unified Memory Architecture (UMA). The system, Synapta, routes each query to K expert adapters and combines their activations inside a RoutedLoRALinear module with a bounding mechanism to prevent representation collapse.

The study proceeds in four pre-registered phases spanning 800+ total real inferences. Key findings:

- On single-domain queries, K=2 composition does NOT outperform single-adapter routing (semantic similarity Δ = −0.011 vs threshold > +0.05, FAIL). The negative result is explained by evaluation design: single-domain benchmarks have no composition signal to recover.
- On genuinely multi-domain queries with oracle routing, K=2 composition yields a directionally positive gain (Δ = +0.0171) that misses the pre-registered +0.03 threshold (FAIL). The effect is highly domain-pair-dependent.
- A confidence-gated embedding router successfully eliminates oracle routing, correctly classifying 100% of single-domain queries as K=1 and enabling K=2 composition autonomously.
- Synapta's autonomous pipeline outperforms Mistral-7B-Instruct on multi-domain expert queries by +5.7% in semantic similarity (0.6525 vs 0.6170) while using 75% less unified memory (~1,100 MB vs ~4,400 MB).

All pre-registered criteria, raw inference logs (800+ entries), and the Mistral-7B verified comparison are reported honestly. The codebase is publicly available at https://github.com/uditjainstjis/mewtwo.

1. Introduction

Low-Rank Adaptation (LoRA) [Hu et al., 2022] enables efficient model specialization without full retraining. Given a frozen base model M and a target domain, LoRA trains compact adapter matrices (A, B) — typically ~20 MB for rank-16 adapters — that inject domain knowledge at inference time. Dozens of specializations can coexist in memory and be hot-swapped per query.

A natural extension is: can multiple adapters be activated simultaneously to handle queries spanning domain boundaries? For example, "How do statistical methods improve diagnostic accuracy in

radiology?" requires both MEDICAL_DIAGNOSIS and MATHEMATICS simultaneously. No single adapter contains the full answer; two adapters must compose to bridge domains.

Apple Silicon Unified Memory Architecture (UMA) makes this question tractable. On UMA, CPU and GPU share the same physical address space, meaning multiple adapter weight matrices can remain memory-mapped without explicit VRAM copy. Multi-adapter coexistence is architecturally natural in a way that it is not on discrete GPU setups.

I investigate this through a pre-registered four-phase empirical study with three success criteria committed to before seeing data:

Phase	Criterion	Metric	Threshold	ID
v1 (SD)	Compositional accuracy	Δ_SIM(AC − SA)	> +0.05	H0
v1 (SD)	Perplexity preservation	PPL(AC) ≤ PPL(SA)	—	—
v1 (SD)	Latency overhead	Δ_LAT	≤ 10%	—
v2 (MD)	SD non-inferiority	Δ_SIM on SD	≥ −0.005	H1
v2 (MD)	MD compositional gain	Δ_SIM(AC-v2 − SA) on MD	> +0.03	H2
v2 (MD)	PPL preservation	PPL(AC-v2) ≤ PPL(SA)	both splits	H3
v2 (MD)	Latency bound	Δ_LAT	≤ 15%	H4
v2 (MD)	Clamp necessity	Δ_SIM(clamped − unclamped)	> 0	H5

The primary contribution is not a winning method — several pre-registered tests fail — but a rigorous, reproducible characterization of what multi-adapter composition can and cannot do on a constrained edge device, and the conditions under which it delivers a measurable intelligence-density advantage over larger monolithic models.

2. Related Work

Mixture-of-LoRA and multi-adapter composition. X-LoRA [Buehler, 2024], MiLoRA, and MALoRA explore gating mechanisms for composing LoRA experts at the token level. These approaches typically operate on server-grade hardware and require per-token weight recomputation. The present study tests coarser prompt-level routing on a memory-constrained UMA device.

Multi-LoRA serving. S-LoRA [Sheng et al., 2023] and Punica [Chen et al., 2023] optimize throughput for serving many single adapters concurrently. Their focus is batching and memory paging rather than semantic composition of two adapters for a single request — the distinct problem studied here.

Activation steering and norm control. Representation engineering [Zou et al., 2023] and activation addition [Turner et al., 2023] demonstrate that steering internal activations is feasible. The norm-proportional clamp proposed here uses the base model's layer-l activation norm ||z_l|| as an energy reference for bounding adapter injection.

On-device LLM inference. MLX [Apple, 2023] provides native Apple Silicon inference with UMA-aware memory management. 4-bit quantization makes 1.5B-class models deployable at <2 GB of unified memory, enabling the multi-adapter coexistence studied here.

Honest reporting of negative results. This paper follows the tradition of pre-registered negative results [Henderson et al., 2018; Dodge et al., 2019] — reporting exactly what the data shows, including hypothesis failures, rather than selecting favorable subsets post-hoc.

3. Method

3.1 Problem Formulation

Let M_θ be a frozen 4-bit quantized language model. Let $E = \{(A_i, B_i)\}$ for $i = 1..N$ be a registry of $N = 20$ domain-specific LoRA adapters (Legal Analysis, Medical Diagnosis, Python, Mathematics, Cryptography, Philosophy, etc.). For a query q, a router selects $K \leq N$ adapters with routing weights $\{w_i\}$. The system composes these adapters within a modified linear layer at every attention projection.

3.2 RoutedLoRALinear

Each nn.Linear and nn.QuantizedLinear layer in the transformer is replaced with a RoutedLoRALinear module. At inference time, the forward pass at layer l is:

$$z_l = W_l \cdot x \qquad \text{(base model output)}$$

$$m_l = \Sigma_{i \in \text{top-}K} \; w_i \cdot (\alpha/r) \cdot (x \cdot A_{i,l}) \cdot B_{i,l} \qquad \text{(adapter injection)}$$

where $\alpha = 16$ is the LoRA scaling factor and $r = 16$ is the adapter rank. All 20 adapter weight matrices are loaded into UMA at startup and remain memory-mapped for zero-copy access.

3.3 Clamping Mechanisms

Unclamped addition causes catastrophic degradation (shown empirically in §4.1). Two clamping strategies were implemented and compared:

Weight-Cap Mode (live backend, v1 and v2): Per-adapter coefficient cap:

$$w_i^* = \min(p_i, c)$$

$$\text{output}_l = z_l + \Sigma_i \; w_i^* \cdot (\alpha/r) \cdot (x \cdot A_{i,l}) \cdot B_{i,l}$$

The clamp hyperparameter $c = 0.5$ was used throughout. This globally caps how much influence any single adapter can have.

Norm-Ratio Mode (implemented in v2b for ablation): Per-layer activation norm gate:

$$\gamma_l = \min(1.0, c \cdot ||z_l||_2 / (||m_l||_2 + \varepsilon))$$

$$\text{output}_l = z_l + \gamma_l \cdot m_l$$

This per-layer scalar gate asks: how large is the combined adapter signal relative to the base model's activation? It attenuates m_l proportionally if it is disproportionately large. $\varepsilon = 1e\text{-}6$ prevents division by zero. Both modes are selectable at runtime via set_clamp_mode() in dynamic_mlx_inference.py.

3.4 Router Architecture

Four routing strategies were studied across the four phases:

CoT Router (Phase 1): The base model itself generates a chain-of-thought classification and outputs a domain tag. Returns a one-hot vector (K=1 only). Achieves ~60% exact-match accuracy. Cannot natively express a soft distribution over two domains.

Oracle Router (Phase 2 ablation): Routing weights are derived from each question's required_adapters metadata, assigning equal weight (0.5) to each of the two correct domains. Isolates the composition question from the routing question: 'does K=2 help when both adapters are correct?'

EmbeddingRouter (Phase 3): Domain centroids are computed over synthetic domain corpora using all-MiniLM-L6-v2 sentence embeddings. Each query is embedded, and the K closest domain centroids by cosine distance determine routing weights. Achieves 78.7% exact-match accuracy. Supports soft multi-label routing with zero training requirement.

GatedRouter (Phase 3 — Champion): Wraps EmbeddingRouter with a confidence gate. Let $p_1 \geq p_2$ be the top-two router probability scores. The gating rule is:

$$K = 1 \quad \text{if} \quad p_1 > 0.5 \quad \text{AND} \quad (p_1 - p_2) > 0.2$$
$$K = 2 \quad \text{otherwise}$$

If the router's top score is dominant and the gap to the second candidate is wide, the query is classified as single-domain. Otherwise, K=2 composition is triggered. This prevents noise injection on single-domain queries while enabling composition on boundary cases.

4. Experiments

4.1 Phase 1: Single-Domain Evaluation (v1)

Setup. Four methods evaluated on 100 single-domain questions (5 per domain × 20 domains) against Qwen2.5-1.5B-Instruct-4bit on Apple M3 Max via MLX. Total inferences: 400.

Benchmark characterization. *~90% of the 100 reference answers reduce to one of two boilerplate templates differing only in the domain noun. This is a single-domain synthetic recall benchmark — not a compositional reasoning benchmark. This characterization is critical for interpreting the negative result below.*

Metrics: Semantic similarity (cosine via all-MiniLM-L6-v2), perplexity of ground-truth text, wall-clock latency.

Method	K	Clamp c	Avg Sim ↑	Avg PPL ↓	Avg Lat (s)
Baseline	0	0.001	0.620	64.5	2.80
SingleAdapter	1	0.5	0.622	60.9	2.69
AdaptiveClamp	2	0.5	0.611	58.0	2.67
UnclampedMix	2	999	0.557	51.2	2.51

Hypothesis verdicts:

Criterion	Measured	Threshold		Verdict
Δ_SIM(AC − SA)	-0.011	> +0.05		FAIL
PPL(AC) vs PPL(SA)	58.0 < 60.9	AC ≤ SA		PASS
Δ_LAT	-0.7%	≤ 10%		PASS

Unclamped mixing is catastrophic. On 8/100 prompts, UnclampedMix produces near-random text (semantic similarity < 0.1). Average drops to 0.557 — Δ = −0.065 relative to SingleAdapter. Norm bounding is structurally necessary.

Composition on single-domain data adds noise, not signal. AdaptiveClamp underperforms SingleAdapter by −0.011. This is the expected result when both adapters are activated for a query that requires only one domain.

Domain-level heterogeneity. AdaptiveClamp outperforms SingleAdapter in 4/20 domains:

Domain	AC	SA	Δ	
MEDICAL_DIAGNOSIS	0.713	0.683	+0.030	
MATHEMATICS	0.587	0.543	+0.044	
QUANTUM_CHEMISTRY	0.634	0.611	+0.023	
PHILOSOPHY	0.617	0.596	+0.021	

But underperforms in 16/20, with particularly large losses on:

Domain	AC	SA	Δ	
MARITIME_LAW	0.464	0.609	−0.145	
ANCIENT_HISTORY	0.516	0.555	−0.039	
SANSKRIT_LINGUISTICS	0.703	0.738	−0.035	
CRYPTOGRAPHY	0.559	0.592	−0.033	

4.2 Phase 2: Multi-Domain Evaluation (v2)

Motivation. The v1 negative result is interpretively ambiguous: single-domain questions have no composition signal to recover. To test whether K=2 helps when both adapters are relevant, I constructed a genuinely multi-domain benchmark.

Multi-domain benchmark (data/multidomain_eval_v2.json). 40 questions, each requiring knowledge from exactly two distinct domains simultaneously (e.g., MARITIME_LAW ×

CRYPTOGRAPHY, MEDICAL_DIAGNOSIS × MATHEMATICS, QUANTUM_CHEMISTRY × CLIMATE_SCIENCE). All 20 domains appear across the question set with 39 distinct domain pairs.

Oracle routing. Because the production CoT router returns a one-hot vector, I use oracle routing for K=2 methods: each question's required_adapters metadata provides the two correct domains with equal weight (0.5). This cleanly isolates composition quality from router quality.

Total inferences: 560 = 140 questions × 4 methods, on both SD (100 q.) and MD (40 q.) splits.

Single-domain split results (sanity check)

*On SD questions, oracle routing produces K=1 — each SD question has exactly one domain in required_adapters, so AC-v2 degenerates to single-adapter.

Method	K	Avg Sim ↑	Avg PPL ↓	Avg Lat (s)
Baseline	0	0.6090	64.5	3.700
SingleAdapter	1	0.6064	60.9	3.571
AdaptiveClamp-v2	1*	0.6058	57.9	3.657
UnclampedMix-v2	1*	0.6041	52.3	3.623

Multi-domain split results — main experiment

Method	K	Avg Sim ↑	Avg PPL ↓	Avg Lat (s)
Baseline	0	0.6473	12.7	4.059
SingleAdapter	1	0.6334	12.7	4.057
AdaptiveClamp-v2	2	0.6505	12.6	4.090
UnclampedMix-v2	2	0.6505	12.6	4.100

Hypothesis verdicts

Hypothesis	Measured	Threshold		Verdict
H1: SD Non-Inferiority	Δ = −0.0006	≥ −0.005		PASS
H2: MD Compositional Gain	Δ = +0.0171	> +0.03		FAIL
H3: PPL Preservation	57.9 < 60.9 (SD); 12.6 < 12.7 (MD)	AC ≤ SA		PASS
H4: Latency Bound	+1.9%	≤ 15%		PASS
H5: Clamp Necessity	Δ = 0.0000	> 0		FAIL

The critical sign flip. The sign of the composition effect reverses between v1 and v2:

- v1 (single-domain): Δ_SIM = −0.011 (composition HURTS)
- v2 (multi-domain): Δ_SIM = +0.017 (composition HELPS)

This is the most scientifically important finding of the study: the v1 negative result does not reflect a fundamental failure of multi-adapter composition — it reflects a mismatch between the evaluation protocol and the method's operating condition. When queries genuinely require two domains, the second adapter contributes real signal rather than noise.

Per-question analysis. The MD composition effect exhibits high variance (σ = 0.059). Strongest gains:

- md_32 (MEDICAL_DIAGNOSIS × MATHEMATICS): +0.303
- md_19 (LATEX_FORMATTING × MATHEMATICS): +0.109
- md_20 (LEGAL_ANALYSIS × CRYPTOGRAPHY): +0.083
- md_02 (MARITIME_LAW × BEHAVIORAL_ECONOMICS): +0.068

Largest losses:

- md_09 (PYTHON_LOGIC × ROBOTICS): −0.125
- md_25 (CLIMATE_SCIENCE × ORGANIC_SYNTHESIS): −0.061
- md_38 (ARCHAIC_ENGLISH × LEGAL_ANALYSIS): −0.029

PYTHON_LOGIC and ROBOTICS both target code-and-control representations, likely competing for overlapping subspaces. MEDICAL_DIAGNOSIS and MATHEMATICS address complementary subspaces, enabling additive composition. Domain-pair orthogonality in adapter weight space appears to be a strong predictor of composition quality.

H5 failure — a methodological note. H5 failed because weight-cap clamping and unclamped mixing are algebraically identical under oracle routing: when p_i = 0.5 and c = 0.5, min(0.5, 0.5) = min(0.5, 999) = 0.5. The per-adapter weight-cap mechanism cannot distinguish clamped from unclamped at this operating point. This is an artifact of the weight-cap implementation, not a finding about the norm-ratio clamp, which is addressed directly in §4.3.

4.3 Phase 2b: Clamp Mechanism Ablation (v2b)

Question. Does the theoretically correct per-layer norm-ratio clamp (γ_l) outperform the simpler per-adapter weight-cap (w_i*) on the MD benchmark?

Setup. 120 inferences: 40 MD questions × 3 methods. The full norm-ratio path was implemented in RoutedLoRALinear.__call__() behind a runtime flag (set_clamp_mode()).

Method	Clamp Mode	Avg Sim ↑	Avg PPL ↓	Avg Lat (s)
SingleAdapter	weight_cap	0.6334	12.7	4.008
AC-v2-WeightCap	weight_cap	0.6505	12.6	4.055
AC-v2-NormRatio	norm_ratio	0.6502	12.6	4.221

Δ_SIM(NormRatio − WeightCap) = −0.0003. Exact same semantic score on 36/40 items; within 0.001 on 37/40.

The theoretically elegant per-layer norm-ratio clamp produces no measurable quality improvement over the simpler weight-cap in this experimental regime. At rank-16, adapter activation magnitudes $||m_l||_2$ are small relative to base model activations $||z_l||_2$ across all layers, so $\gamma_l \approx 1.0$ everywhere and the clamp is effectively inactive. The weight-cap is the binding constraint. The norm-ratio may become relevant at higher adapter ranks or higher routing weights.

The norm-ratio mode also adds +4% latency (4.221s vs 4.055s) without quality benefit.

4.4 Phase 2c: Routing Gap Analysis (v2c)

Question. Oracle routing is an upper bound requiring ground-truth domain labels. How much compositional gain does the real multi-label router recover from this ceiling?

Setup. I extended the CoT orchestrator with a cascading top-2 extraction strategy (two sequential generation calls). 120 inferences: 40 MD questions × 3 methods.

Method	Routing	Avg Sim ↑	Avg K	Δ vs SA
SingleAdapter	CoT K=1	0.6296	1.00	—
AC-v2-RealRouter	CoT Top-2	0.6350	1.75	+0.0054
AC-v2-Oracle	Oracle K=2	0.6502	2.00	+0.0206

- Oracle headroom over SingleAdapter: +0.0206
- Realized gain with real router: +0.0054
- Real router recovers 26% of oracle headroom
- Routing gap (Oracle − RealRouter): −0.0152
- Real router achieved K=2 on only 30/40 questions (75%)

Crucially, oracle headroom itself is small (+0.0206) — even perfect routing would not cross the pre-registered +0.03 threshold. The paper cannot attribute the H2 failure entirely to router inadequacy. The binding constraint is the composition mechanism itself — adapter signal magnitude, subspace alignment, or base model capacity at 1.5B parameters.

4.5 Phase 3: Autonomous Confidence-Gated Routing

Question. Can a fully autonomous pipeline — no oracle labels — correctly gate between K=1 and K=2, preserving single-domain performance while recovering multi-domain composition gains?

Router comparison (40 MD queries):

Router	Avg Routing Acc (Exact Match)	Avg Similarity	Avg Latency
Oracle (upper bound)	100.0%	0.6505	4.03s
EmbeddingRouter	78.7%	0.6521	4.05s

ClassifierRouter	78.7%	0.6441	4.07s
MultiLabelCoT	48.7%	0.6431	4.05s

CoT generative multi-label routing fails on 1.5B models. The model produces hallucinations when asked to output a JSON array of relevant domains across 20 classes, achieving only 48.7% exact-match accuracy.

Lightweight discriminative routers succeed. EmbeddingRouter (cosine similarity to domain centroids) and ClassifierRouter (logistic regression) both achieve 78.7% accuracy at negligible overhead. EmbeddingRouter was selected as champion for its interpretability and zero training requirement.

GatedRouter on mixed queries (140 total: 100 SD + 40 MD):

Single-domain split (100 queries): Classification accuracy (K=1 expected): 100.0% · Avg K used: 1.00 · Avg similarity: 0.6058

Multi-domain split (40 queries): K=2 activation rate: 12.5% · Avg K used: 1.12 · Avg similarity: 0.6525

The GatedRouter achieved perfect single-domain isolation: no single-domain query received unwanted second-adapter injection. On multi-domain queries, the router predominantly assigned K=1 (87.5%), recognizing that a single dominant domain often suffices. When it did activate K=2, the similarity result (0.6525) matched or exceeded oracle K=2 performance.

4.6 Phase 4: Mistral-7B Intelligence Density Verification

Question. Can a 1.5B model with expert adapter routing outperform an unspecialized 7B model on multi-domain expert queries?

Setup. The exact same 40 MD hard questions were run against Mistral-7B-Instruct-v0.3-4bit via local Ollama backend. Synapta used the Phase 3 GatedRouter. Raw results are archived in results/mistral_md_results.json and the comparison is formally verified in results/mistral_vs_synapta_verified.md.

Metric	Mistral-7B (Generalized)	Synapta-Gated (Virtual MoE)	Δ
MD Avg Semantic Similarity	0.6170	0.6525	+5.7%
Unified Memory Footprint	~4,400 MB	~1,100 MB	−75%
Avg Latency per Query	~9.20s	~4.05s	~2.3× faster
K=2 Activation Rate	N/A	12.5%	—

On the 40 multi-domain expert queries in this benchmark, Synapta's autonomous Virtual MoE pipeline outperforms a model with 4.6× more parameters by 5.7% in semantic similarity, at 25% of the memory footprint and less than half the latency.

Scope caveat. *This comparison is specific to the 40-question multi-domain benchmark for which Synapta's adapters were explicitly trained. It is not a general claim about Synapta's superiority over Mistral-7B across all tasks.*

5. Discussion

5.1 The Evaluation Dependency of Composition Effects

The central lesson is that multi-adapter composition effects are strongly evaluation-dependent. The sign of the composition signal reversed completely between v1 (single-domain, −0.011) and v2 (multi-domain, +0.017). A researcher using only the v1 benchmark would conclude that prompt-level composition is harmful; one using only the v2 oracle results might conclude it is beneficial. Both conclusions would be incomplete.

The correct framing is conditional: K=2 composition helps specifically when (a) the query genuinely requires two domains, and (b) both adapters are correctly identified. It hurts when either condition is violated.

5.2 Domain-Pair Orthogonality as a Predictor

Per-question variance on the MD benchmark ($\sigma = 0.059$) reveals that adapter composition quality is domain-pair-dependent. The largest gains (MEDICAL_DIAGNOSIS × MATHEMATICS, +0.303) involve domains that likely occupy orthogonal representational subspaces. The largest losses (PYTHON_LOGIC × ROBOTICS, −0.125) involve domains competing for shared code-and-control representations.

This suggests that adapter subspace orthogonality should be a pre-condition for enabling K=2 composition, rather than a post-hoc explanation. A practical extension would measure pairwise adapter interference (via gradient inner products or activation cosine distances) before committing to multi-adapter injection for a given domain pair.

5.3 Clamp Mechanism Inactivity

The norm-ratio clamp was theoretically motivated as a principled solution to adapter signal magnitude imbalance. However, the v2b ablation shows it is effectively inactive at rank-16 with routing weights of 0.5: adapter activations are small relative to base model activations, so $\gamma_l \approx 1.0$ at every layer and the bound never binds.

The clamp's importance is empirically demonstrated only by comparing against the fully unclamped baseline — not by comparing the two clamped variants. This should be understood as a useful boundary condition: the clamp is necessary for stability at high routing weights or high rank, but is not the limiting factor for quality at the operating parameters studied here.

5.4 Routing as Necessary but Insufficient

The v2c routing gap analysis shows the real router recovers only 26% of oracle headroom. However, oracle headroom itself is only +0.0206 — below the pre-registered +0.03 threshold. Improving routing accuracy to 100% would still not pass H2. The binding constraint is the composition mechanism itself, not router quality. Future work aiming to cross the threshold should focus on composition architecture improvements rather than routing improvements.

5.5 Intelligence Density Under Specialization

The Phase 4 result — +5.7% semantic similarity at 75% lower memory and 2.3× lower latency — is interpretable as a specialization premium: expert adapters inject domain-specific knowledge that a generalist 7B model, trained on broad corpora, has diluted. On the specific 40 expert queries in this benchmark, information relevant to the correct answer is more concentrated in a 1.5B + adapter system than in a generic 7B model.

6. Limitations

- Benchmark scale: The MD benchmark contains 40 questions with uneven domain-pair coverage. Domain-pair interaction claims should be treated as pilot evidence, not statistically robust findings.
- Synthetic training data: All 20 adapters were trained on synthetically generated corpora. The generalization gap to human-authored domain questions is unknown.
- Evaluation metric: Semantic similarity via all-MiniLM-L6-v2 is sensitive to lexical overlap and may not capture logical correctness. The v1 benchmark's template-derived answers make it particularly brittle.
- Single base model: All experiments use Qwen2.5-1.5B-Instruct-4bit. Whether composition effects generalize to other architectures or adapter ranks is not established.
- Mistral comparison scope: The Phase 4 comparison covers only the 40 multi-domain questions for which Synapta's adapters were explicitly trained. Performance differences on general-purpose benchmarks would likely favor Mistral-7B.

7. Conclusion

This paper presents a four-phase, pre-registered empirical study of prompt-level multi-adapter LoRA composition on Apple Silicon UMA. The findings do not support a simple 'composition works' or 'composition fails' narrative — they identify the conditions under which each is true.

On single-domain queries, K=2 composition is consistently harmful (−0.011 relative to K=1). On genuinely multi-domain queries with correct routing, K=2 composition yields a small but consistent gain (+0.017) that is directionally meaningful but misses the pre-registered threshold, and is highly domain-pair-dependent. The norm-ratio clamp and the simpler weight-cap produce identical results at the operating parameters studied. Routing is a bottleneck, but not the primary one: even oracle routing yields only +0.020, which still misses the pre-registered target.

The fully autonomous pipeline — EmbeddingRouter + GatedRouter — successfully eliminates oracle routing by correctly classifying 100% of single-domain queries as K=1, preventing noise injection, while enabling K=2 composition on boundary cases. This system, running within ~1.1 GB of unified memory, outperforms Mistral-7B on the 40 multi-domain expert queries studied by +5.7% in semantic similarity at 2.3× lower latency.

The open research question is whether the composition signal grows at higher base model capacity, higher adapter rank, or with domain-pair orthogonality pre-filtering. All data, benchmarks, and inference infrastructure are publicly released to facilitate follow-up work.

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Appendix A: Pre-Registered Hypothesis Summary

ID	Hypothesis	Measured	Threshold	Phase	Verdict
H0	Compositional accuracy (SD)	Δ = −0.011	> +0.05	v1	FAIL
—	Perplexity preservation	58.0 < 60.9	AC ≤ SA	v1	PASS
—	Latency overhead	−0.7%	≤ 10%	v1	PASS
H1	SD non-inferiority	Δ = −0.0006	≥ −0.005	v2	PASS
H2	MD compositional gain	Δ = +0.0171	> +0.03	v2	FAIL
H3	PPL preservation	SD: 57.9<60.9; MD: 12.6<12.7	AC ≤ SA	v2	PASS
H4	Latency bound	+1.9%	≤ 15%	v2	PASS
H5	Clamp necessity	Δ = 0.0000	> 0	v2	FAIL

Appendix B: Total Inference Count

Phase	Script	Inferences
v1 (SD benchmark)	src/eval/run_eval.py	400
v2 (SD+MD)	src/eval/run_eval_v2.py --real --split both	560

v2b (clamp ablation)	src/eval/run_eval_v2b.py --phase clamp	120
v2c (routing gap)	src/eval/run_eval_v2b.py --phase routing	120
Phase 3 (router comparison)	src/eval/router_ablation.py	120
Phase 3 (gated routing)	src/eval/gated_routing_eval.py	140
Phase 4 (Mistral comparison)	backend/mistral_comparison.py	140
Total		1,600

Appendix C: Reproducibility

Prerequisites: Apple Silicon Mac (M1/M2/M3), ≥16 GB unified memory, Python 3.11+

```
pip install mlx mlx-lm sentence-transformers safetensors scikit-learn

# Phase 1 (v1): 400 inferences, ~30 min
PYTHONUNBUFFERED=1 python3 src/eval/run_eval.py --real

# Phase 2 (v2): 560 inferences, ~41 min
PYTHONUNBUFFERED=1 python3 src/eval/run_eval_v2.py --real --split both

# Phase 2b: 120 inferences, ~10 min
PYTHONUNBUFFERED=1 python3 src/eval/run_eval_v2b.py --phase clamp --real

# Phase 2c: 120 inferences, ~10 min
PYTHONUNBUFFERED=1 python3 src/eval/run_eval_v2b.py --phase routing --real

# Verify result file sizes
wc -l results/v2_both_raw.jsonl           # 560
wc -l results/v2_md_clamp_ablation.jsonl  # 120
wc -l results/v2_md_routing_ablation.jsonl # 120
```